

Question	Marking Guidance	Mark	Comments
02.1	1. No/fewer calcium <u>ions</u> enter synaptic knob OR No/less calcium enter synaptic knob via calcium <u>ion</u> channels; 2. No/fewer synaptic vesicles move to/fuse with presynaptic membrane and no/less glutamate is released; 3. No/less glutamate <u>diffuses</u> across (synaptic cleft); 4. No/less (glutamate attaches) to <u>receptors</u> on the postsynaptic <u>membrane</u>; 5. No/fewer sodium <u>ions</u> enter (postsynaptic neurone) so no/ fewer impulses (sent to brain);	5 (2 x AO1, 3 x AO2)	Note: If no/fewer/less is correctly included for any mark point allow for all other mark points. A description of synaptic transmission without no/less/fewer gains four marks max. 1. Accept $\text{Ca}^{2+}/\text{Ca}^{++}$ 1. Accept 'presynaptic neurone/knob' for synaptic knob but not 'presynaptic membrane' on its own. 2, 3 and 4. Accept acetylcholine or neurotransmitter for glutamate. 5. Accept Na^{+}. 5. Accept 'no transmission' ('of impulses' is in passage) or 'no depolarisation' or 'no action/generator potentials' for fewer impulses, but reject 'messages' and 'signals'. 5. Accept 'threshold (potential) not being reached' for fewer impulses.

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02.2	1. Rapidly/quickly/directly reaches spinal cord; 2. Is broken down by enzymes OR Is broken down by acid OR Is too large to be absorbed;	2 (2 x AO2)	1. Accept 'synapses' for spinal cord. 1. Accept spinal cord and brain but reject brain on its own. 2. Accept denatured by acid. 2. Accept 'not affected by acid/enzymes' 2. Reject broken down by named enzymes which are not proteases. 2. Ignore location of enzymes. 2. Accept description for 'absorption' e.g. 'cross the ileum'.

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02.3	Correct answer of $5/4.7(232) \times 10^{-5} = \mathbf{2 \text{ marks}};$ Answer of 5 or 47(232) with incorrect order of magnitude = 1 mark OR Answer of $2/1.9(68) \times 10^{-6}$ (i.e., not multiplied by 24) = 1 mark OR Answer of $6/5.8/5.76 \times 10^{-7}$ (i.e., not multiplied by 82) = 1 mark OR Answer of $2/1.6/1.57(44) \times 10^{-5}$ (i.e., not multiplied by 3) = 1 mark OR Answer of $6/5.9/5.904 \times 10^{-6}$ (i.e., not multiplied by 8) = 1 mark;	2 (2 x AO2)	Accept any equivalent numerical representation of the correct answer. Ignore numbers after 4.7 and 47.

Question	Marking Guidance	Mark	Comments
02.4	1. (Sensation of) pain is subjective OR Pain sensitivity/threshold/tolerance varies; 2. To ensure that differences (in pain detection) were (statistically) significant OR For (valid) comparison;	2 (2 x AO3)	1. Accept 'could exaggerate pain'. 2. Accept 'not due to chance' for significant. 2. Do not credit 'to determine if results are significant.'